COURSE SLIDES · WSQ

Robotics Process Automation (RPA) for Beginners

Course Code: TGS-2026064721 ·
2-Day Hands-On Training
Version v15.0 · 23 August 2026



Tertiary Infotech Academy Pte Ltd
UEN: 201200696W

COURSE PURPOSE

Welcome

Build reliable UiPath automations that select the right work, express clear logic, recover from failure and remain manageable in production.

MANDATORY

Digital Attendance

AM

Scan and submit morning attendance.

PM

Scan and submit afternoon attendance.

Assessment

Complete assessment attendance and TRAQOM.

Requirement

At least 75% attendance is required.

TRAINER

Meet Your Trainer

Name

Dr. Alfred Ang

Organisation

Tertiary Infotech Academy Pte Ltd

Role

Principal trainer and facilitator

Contact

enquiry@tertiaryinfotech.com

ICEBREAKER

Learner Introductions

Who

Your name, role and organisation.

Experience

Any process, scripting or automation exposure.

Opportunity

One repetitive task you would like to improve.

Success

What a useful outcome looks like for you.

WORKING AGREEMENT

Ground Rules

01

Participate actively and ask questions.

02

Keep devices silent during learning.

03

Respect confidentiality in examples.

04

Return punctually from breaks.

05

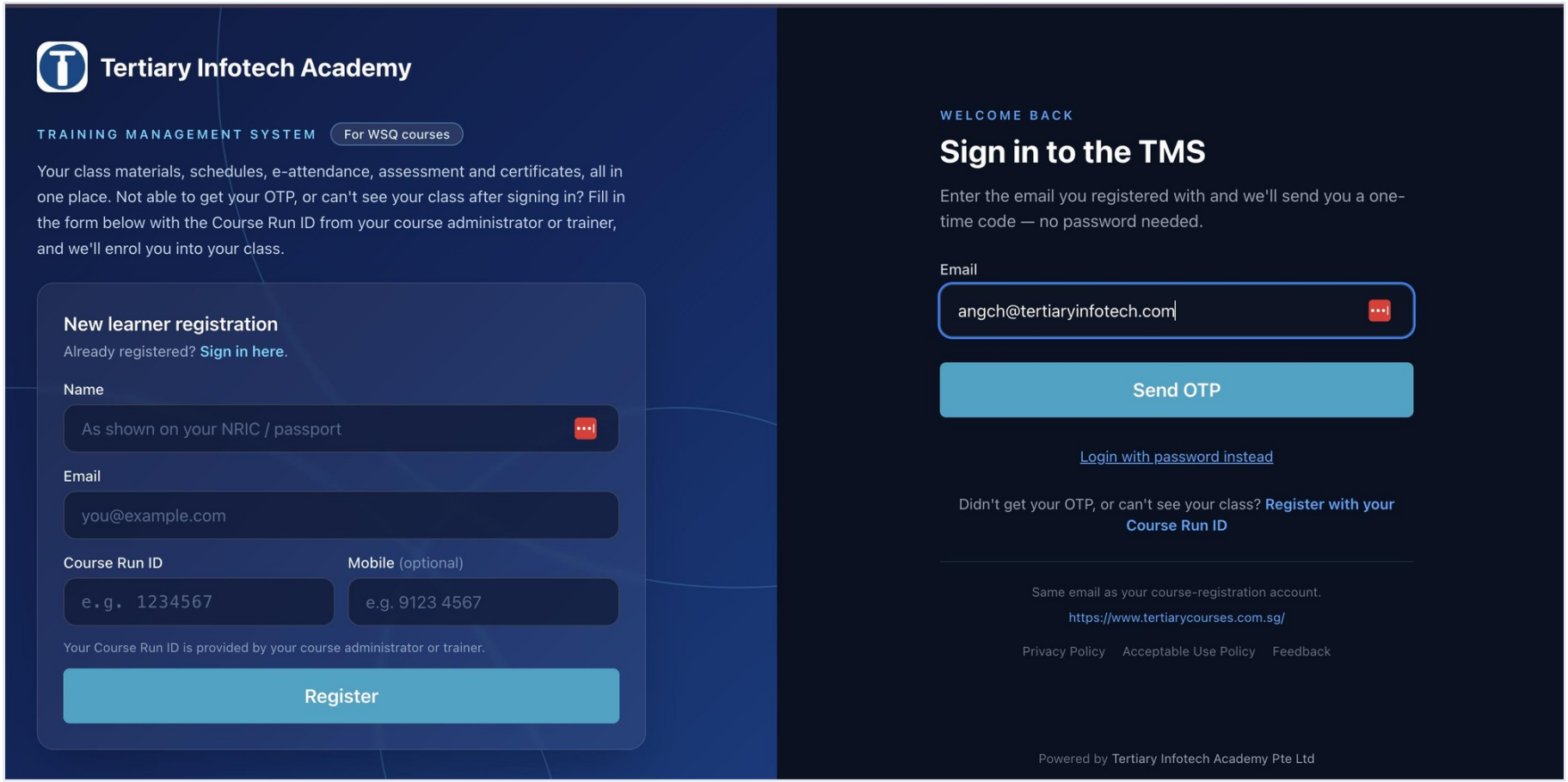
Use only mock data in labs.

06

Capture evidence before moving on.

COURSE PORTAL

Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

LEARNER RESOURCES

Access Course Material

1. Sign in to LMS
2. Open this course
3. Download slides and guide
4. Open lab repository
5. Retain your evidence

INSIGHT The Learner Guide and individual lab folders contain the detailed procedures.

COURSE MAP

Two-Day Learning Journey

Day 1

RPA foundations, workflow design and data-driven automation

Day 2

UI automation, resilience, deployment and assessment

Practice

Ten connected UiPath labs with mock business data.

Assessment

13 written questions and five practical tasks.

BY THE END

Learning Outcomes

LO1

Determine appropriate RPA opportunities and best practices for attended and unattended automation.

LO2

Plan and construct UiPath workflows using sequences, flowcharts, state machines, variables, arguments and control flow.

LO3

Design and validate UI, web, data and Excel automation processes with reliable selectors and structured outputs.

LO4

Test, debug, log and handle exceptions in RPA workflows.

LO5

Package, deploy, monitor and optimise UiPath processes using Assistant and Orchestrator.

TSC Abilities

A1

Determine best practices for building autonomous software robots

A2

Plan processes used to design and build software robots

A3

Define technical requirements for software robots

A4

Construct fully functional software robots

A5

Validate the design of software robots

A6

Introduce suitable practices in robot design and planning

A7

Gather feedback on robot performance

A8

Review coding, testing and design criteria

A9

Optimise robot performance during and after deployment

KNOWLEDGE EVIDENCE

TSC Knowledge

K1
Autonomy, agency, and attended versus unattended operation

K2
Automating repetitive tasks and intelligent process automation

K3
Single-agent and multiple-agent robot patterns

K4
Sequences, flowcharts, and state-machine workflows

K5
User-interface automation and recording

K6
System activities, user events, and loop selection

K7
Recording and extracting structured data from websites

K8
Testing, debugging, logging, and exception handling

K9
Multi-robot management and Orchestrator

K10
Industry best practices for software robots

K11
Application and data integration

K12
Packaging, embedding, deploying, and running robots

K13
Validation, verification, and resilient exception handling

Assessment Components

WRITTEN ASSESSMENT

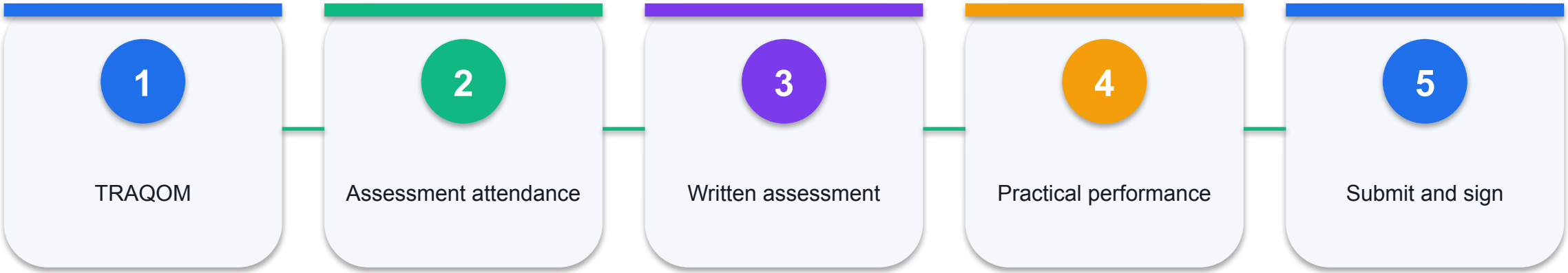
- Written Assessment (WA / SAQ) — 13 open-ended questions, 60 minutes, open book.
- 13 questions map one-to-one to K1–K13.
- Answer in your own words using course concepts.

PRACTICAL PERFORMANCE

- Practical Performance (PP) — 5 connected UiPath tasks, 90 minutes, open book.
- Five connected tasks map to A1–A9.
- Retain the requested UiPath and output evidence.

FINAL SESSION

Assessment Flow



INSIGHT Candidates must achieve at least 75% attendance and be assessed Competent (C).

PROCEDURE BOUNDARY

Detailed procedures live outside this deck

Use the Learner Guide for complete procedures, troubleshooting and evidence criteria. Use each lab folder for its UiPath starter project, CSV and Excel mock data.

TOPIC 01

01

Introduction to RPA

Opportunity selection · attended and unattended automation · value · governance · intelligent automation

LO1 · A1/A6 · K1/K2/K3/K10

Topic 01 · Concept Map

RPA operates the interface

A software robot performs rule-based actions through applications much as a user would.

Choose stable, repetitive work

Volume, clear rules, structured data and low exception rates make strong candidates.

Attended supports people

The user starts or assists the automation at the desktop.

Unattended runs centrally

Orchestrator schedules and monitors work without a user at the screen.

IPA expands the boundary

OCR, document understanding and AI handle variability beyond simple deterministic rules.

Automation needs controls

Ownership, credentials, logging, exception paths and human review make robots sustainable.

Benefits of RPA



Accuracy

Consistent execution reduces manual transposition and missed-step errors.

Capacity

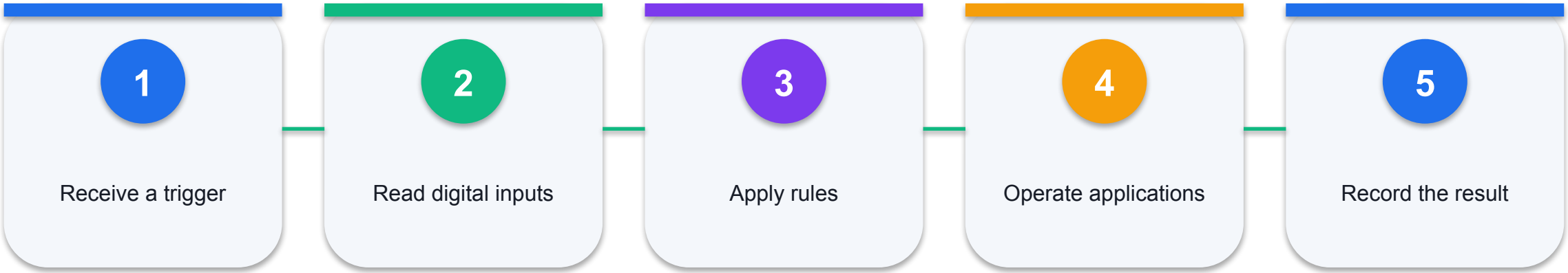
Digital workers extend throughput without making people repeat low-value actions.

Traceability

Logs create evidence for audit, support and continuous improvement.

WORKFLOW

How an RPA Robot Delivers Work



INSIGHT The robot is deterministic: clear inputs, rules and exception paths matter more than human-like appearance.

DECISION VIEW

Attended vs Unattended Automation

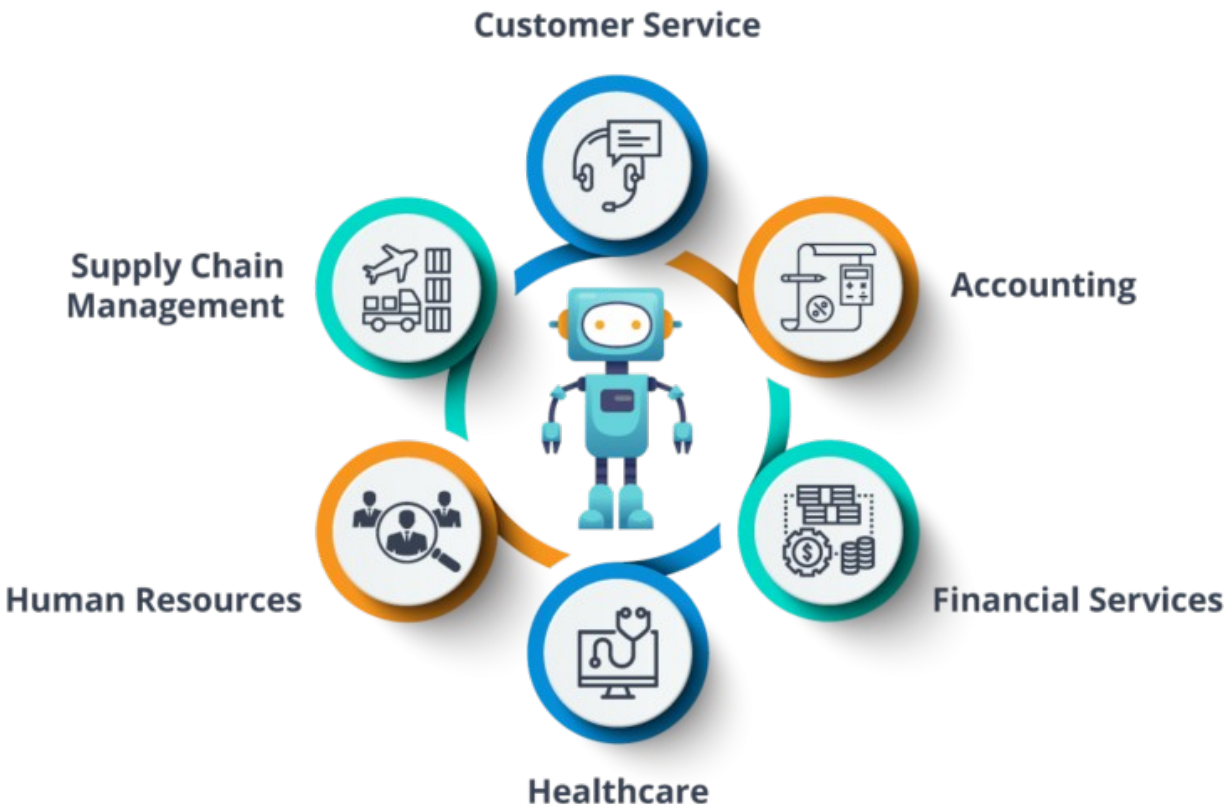
ATTENDED

- User starts the automation
- Runs in the user session
- Supports front-office decisions
- Human handles judgement

UNATTENDED

- Trigger or schedule starts it
- Runs on managed capacity
- Processes back-office volume
- Orchestrator monitors execution

Where RPA Creates Value



Front office

Customer service and healthcare use attended assistance.

Back office

Accounting, finance and supply-chain work suits unattended processing.

Shared pattern

Structured inputs, stable rules and digital applications.

PRACTITIONER VIEW

The RPA Value Proposition

Speed

Shorter cycle time · 24/7 execution · Rapid queue clearance

Quality

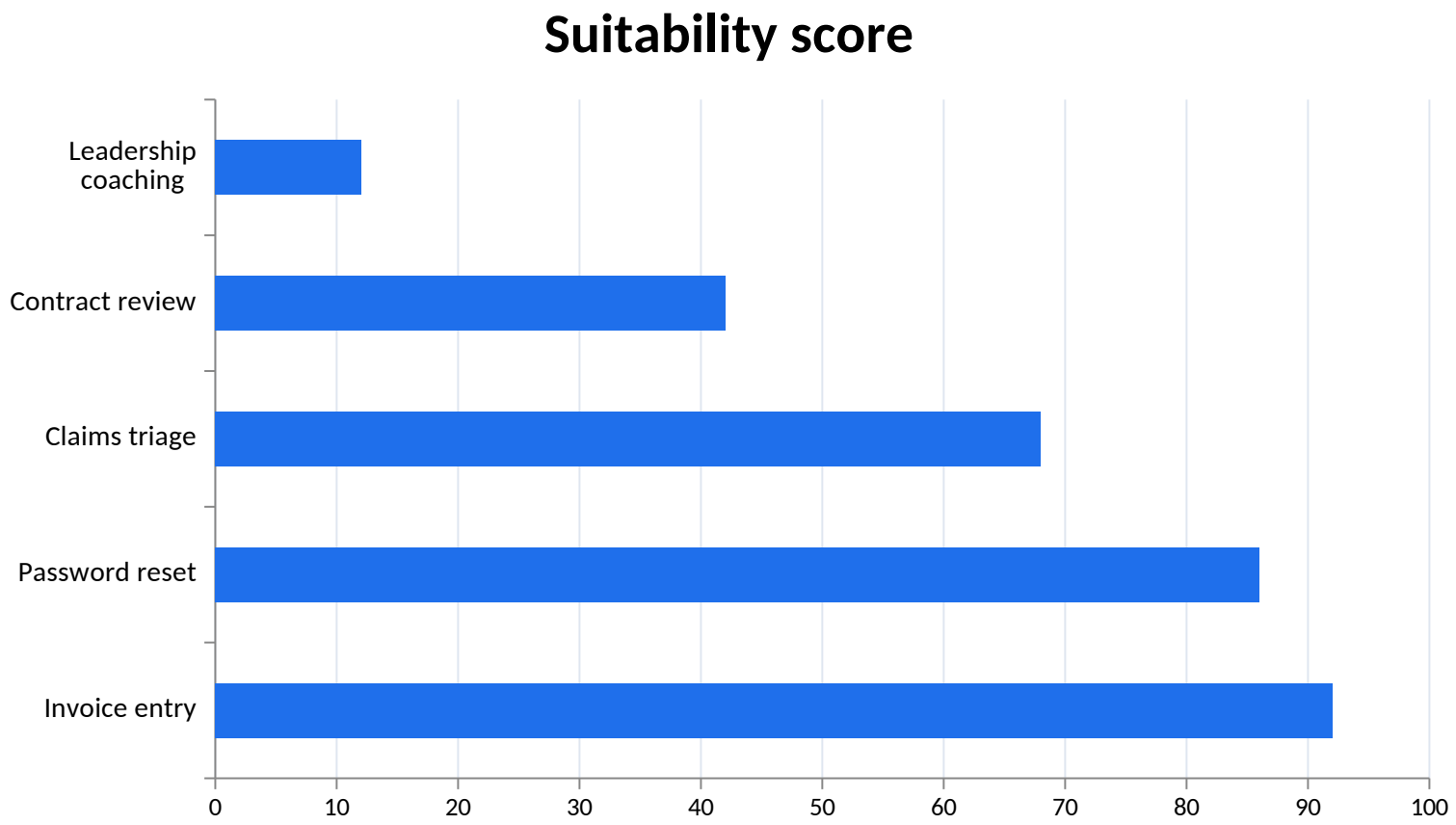
Repeatable rules · Fewer manual errors · Traceable outcomes

People

Less repetitive work · More exception handling · More customer focus

EDITABLE POWERPOINT CHART

Illustrative Automation Suitability



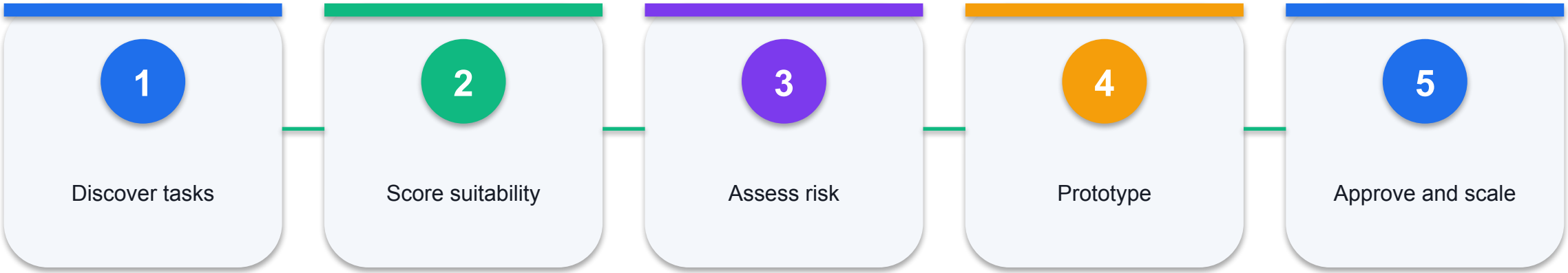
WHAT TO NOTICE

High-volume, rule-based and digitally accessible work scores highest. Indicative teaching figures.

Indicative teaching data

WORKFLOW

Automation Opportunity Funnel



INSIGHT A promising idea is not a production robot until risk, ownership and evidence are clear.

DECISION VIEW

RPA vs API Integration

RPA

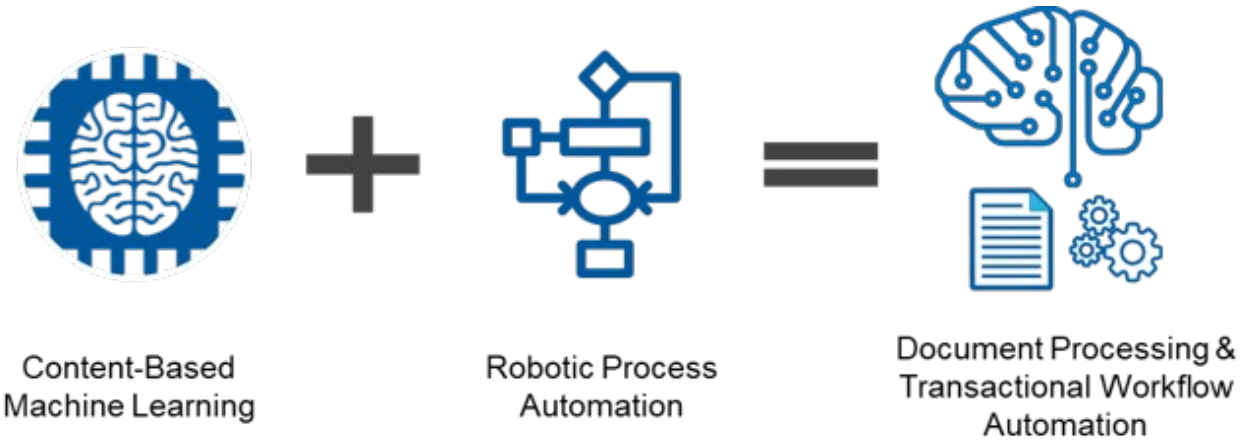
- Operates the existing user interface
- Fast when no API is available
- Sensitive to UI changes
- Good bridge or tactical layer

API

- Connects through a defined contract
- Usually faster and more stable
- Requires an available, governed API
- Best for strategic system integration

LEGACY CONTENT · REBUILT

RPA and Intelligent Automation



RPA

Executes deterministic interface and data actions.

AI capability

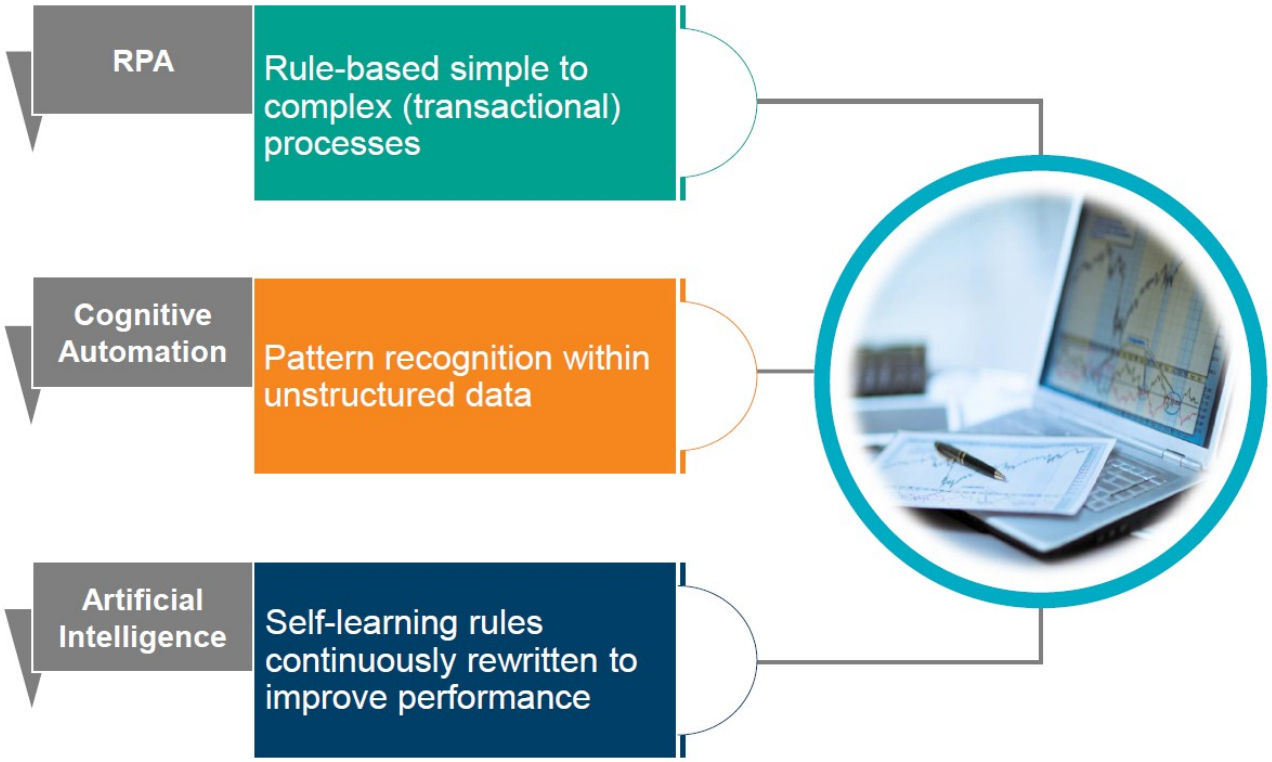
Classifies, extracts or predicts where rules alone are insufficient.

Combined system

A controlled workflow applies AI output with thresholds and human review.

LEGACY CONTENT · REBUILT

The Intelligence Continuum



Rule based

Stable inputs and explicit decisions.

Cognitive

Pattern recognition across less-structured data.

Learning

Models may change behaviour and therefore need monitoring.

PRACTITIONER VIEW

Single and Multiple Robot Patterns

Single robot

One queue or task · Simple ownership · Limited scale

Multiple robots

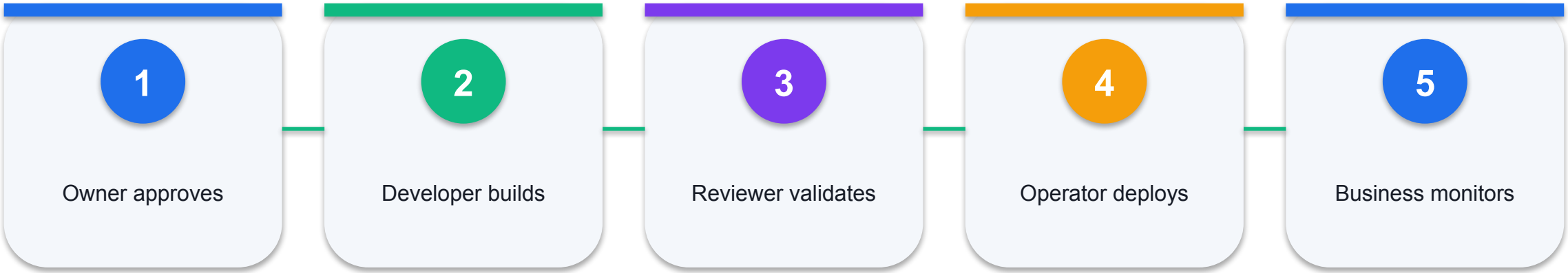
Parallel capacity · Shared queues · Central coordination

Control plane

Schedules and assets · Credential isolation · Unified monitoring

WORKFLOW

RPA Governance Lifecycle



INSIGHT Separation of duties reduces the risk that one person designs, approves and releases uncontrolled automation.

DECISION VIEW

Good and Poor RPA Candidates

STRONG CANDIDATE

- Rules are explicit
- Inputs are digital and structured
- Volume is meaningful
- Exceptions are understood

WEAK CANDIDATE

- Frequent interface redesign
- Tacit judgement dominates
- Unstable or missing data
- No process owner

PRACTITIONER VIEW

RPA Business Case Measures

Baseline

Volume · Handle time · Error and rework rate

Benefit

Hours released · Cycle-time reduction · Avoided defects

Cost and risk

Build and licences · Support effort · Control exposure

TOPIC 01 · HANDS-ON

Lab 1 · Assess an RPA Opportunity and Build a First Sequence

Score a small set of workplace processes, select a strong candidate, then run a starter UiPath sequence that reads the selected process and writes an evidence summary.

1. Inspect candidates
2. Score rules and volume
3. Select best fit
4. Run starter sequence
5. Capture evidence

DELIVERABLE A scored opportunity register, a selected automation candidate and a runnable UiPath sequence

Tools: UiPath Studio Community, CSV, Excel, Write Line · Detailed procedure: Learner Guide and labs/lab-01-*/README.md

EVIDENCE · A1, A6 · K1, K2, K10

Lab 1 · Verification

EXPECTED RESULT

The workbook ranks a rule-based digital process highest, and Main.xaml runs successfully with a clear selected-process message in the Output panel.

TROUBLESHOOTING

If the project opens with missing dependencies, choose Manage Packages > Restore. If the output is blank, inspect the selectedProcess variable scope and default value.

EVIDENCE TO RETAIN

Completed XLSX plus one screenshot showing the successful UiPath Output message.

TOPIC 01 · HANDS-ON

Lab 2 · Model Attended and Unattended Service Automation

Classify service requests and design two operating modes: an attended desktop assistant for judgement-heavy cases and an unattended route for complete, low-risk requests.

1. Read requests
2. Check completeness
3. Assess risk
4. Choose operating mode
5. Summarise routing

DELIVERABLE A routing matrix and UiPath workflow that labels each request Attended, Unattended or Human Review

Tools: UiPath Studio Community, CSV, Excel, If, Switch, Write Line · Detailed procedure: Learner Guide and labs/lab-02-*/README.md

EVIDENCE · A1, A6 · K1, K3, K10

Lab 2 · Verification

EXPECTED RESULT

Three test cases reach three intended routes, every route states a reason, and incomplete requests never continue to unattended processing.

TROUBLESHOOTING

If all cases take the same branch, check Boolean values and remove surrounding spaces. If Switch misses a case, verify text casing or normalise with Trim.ToLower.

EVIDENCE TO RETAIN

Routing matrix and screenshots of the three distinct UiPath outcomes.

CHECK YOUR UNDERSTANDING

Topic 01 · Recap

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Unattended runs centrally

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IPA expands the boundary

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Automation needs controls

Ownership, credentials, logging, exception paths and human review make robots sustainable.

TOPIC 02

02

RPA Workflows

Sequences · flowcharts · state machines · variables · arguments · decisions · loops · data tables

LO2 · A2/A3/A4 · K4/K6/K11

Topic 02 · Concept Map

Sequence

Best for a short linear path with a clear beginning and end.

Flowchart

Makes branching logic and alternate paths easy to inspect.

State machine

Models event-driven work that moves between named states.

Variables hold local state

Name, type, default value and scope determine what a value can do.

Arguments cross boundaries

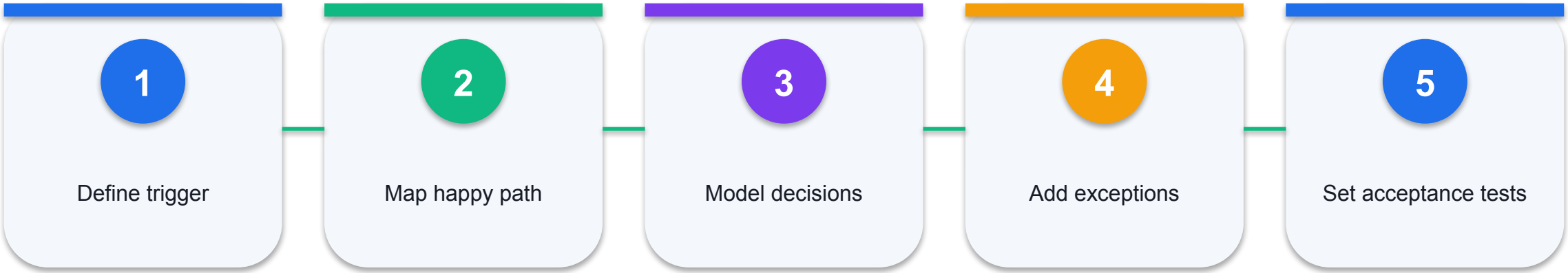
In, Out and In/Out arguments create explicit contracts between workflows.

Control flow expresses rules

If, Switch, For Each, While and Do While turn requirements into executable logic.

WORKFLOW

From Requirement to Workflow



INSIGHT A workflow is executable process design, not a recording of random clicks.

PRACTITIONER VIEW

Three Workflow Containers

Sequence

Linear · Compact · Reusable block

Flowchart

Branching · Visual logic · Multiple paths

State machine

Events · States · Long-running cases

DECISION VIEW

Sequence vs Flowchart

SEQUENCE

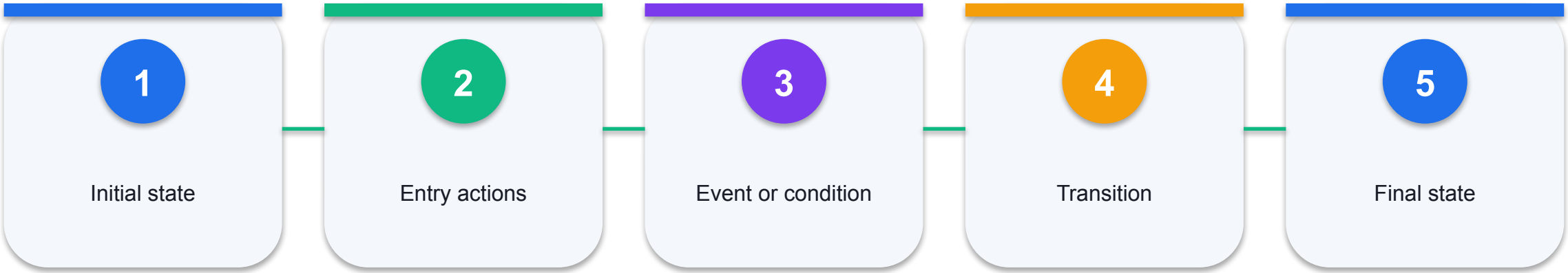
- Reads top to bottom
- Best for a straight path
- Easy to invoke as a component
- Can become hard to scan when branches grow

FLOWCHART

- Exposes branches visually
- Best for decision-heavy work
- Supports loops and alternate paths
- Needs disciplined labels

WORKFLOW

State Machine Anatomy



INSIGHT State machines are useful when events determine what happens next, not merely the next line of work.

PRACTITIONER VIEW

Variable Design

Name

Meaningful noun · Consistent casing · Avoid generic temp

Type

String and Boolean · Numbers and DateTime · DataTable and collections

Scope

Smallest useful scope · Avoid accidental coupling · Dispose sensitive data

DECISION VIEW

Variables vs Arguments

VARIABLE

- Store data inside one workflow
- Scope controls local visibility
- Good for intermediate state
- Not a cross-workflow contract

ARGUMENT

- Pass data between workflows
- Direction is explicit
- Supports reusable components
- Document type and meaning

PRACTITIONER VIEW

Argument Directions

In

Caller supplies value · Read inside workflow · Configuration or input

Out

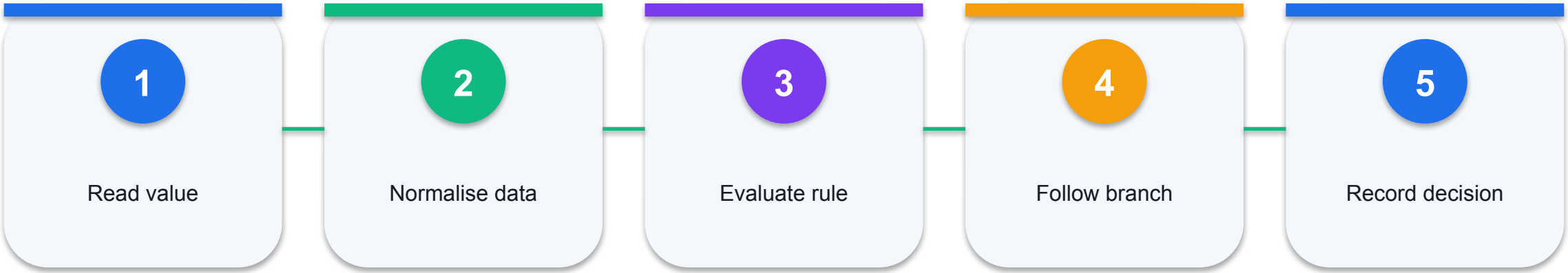
Workflow returns value · Written inside workflow · Result or evidence

In/Out

Caller supplies value · Workflow changes it · Use sparingly

WORKFLOW

Decision Logic Pattern



INSIGHT Normalising once before a decision makes conditions easier to read and test.

DECISION VIEW

If vs Switch

IF

- Binary true/false branch
- Any Boolean expression
- Ideal for thresholds
- Nested Ifs become hard to scan

SWITCH

- One value, many cases
- Clear categorical routing
- Default branch is essential
- Cases must be mutually understood

PRACTITIONER VIEW

Choosing a Loop

For Each

Known collection · One item at a time · Most business tables

While

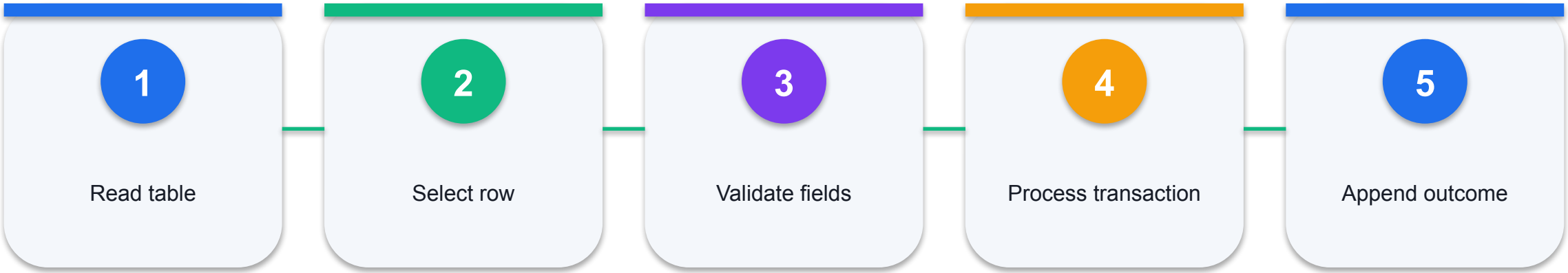
Condition checked first · May run zero times · Guard against infinite loops

Do While

Runs at least once · Condition checked last · Useful for retries/prompts

WORKFLOW

For Each Row Transaction Pattern



INSIGHT Keep the row result with the row identifier so every transaction is traceable.

PRACTITIONER VIEW

DataTable Essentials

Schema

Named columns · Correct data types · Required fields

Rows

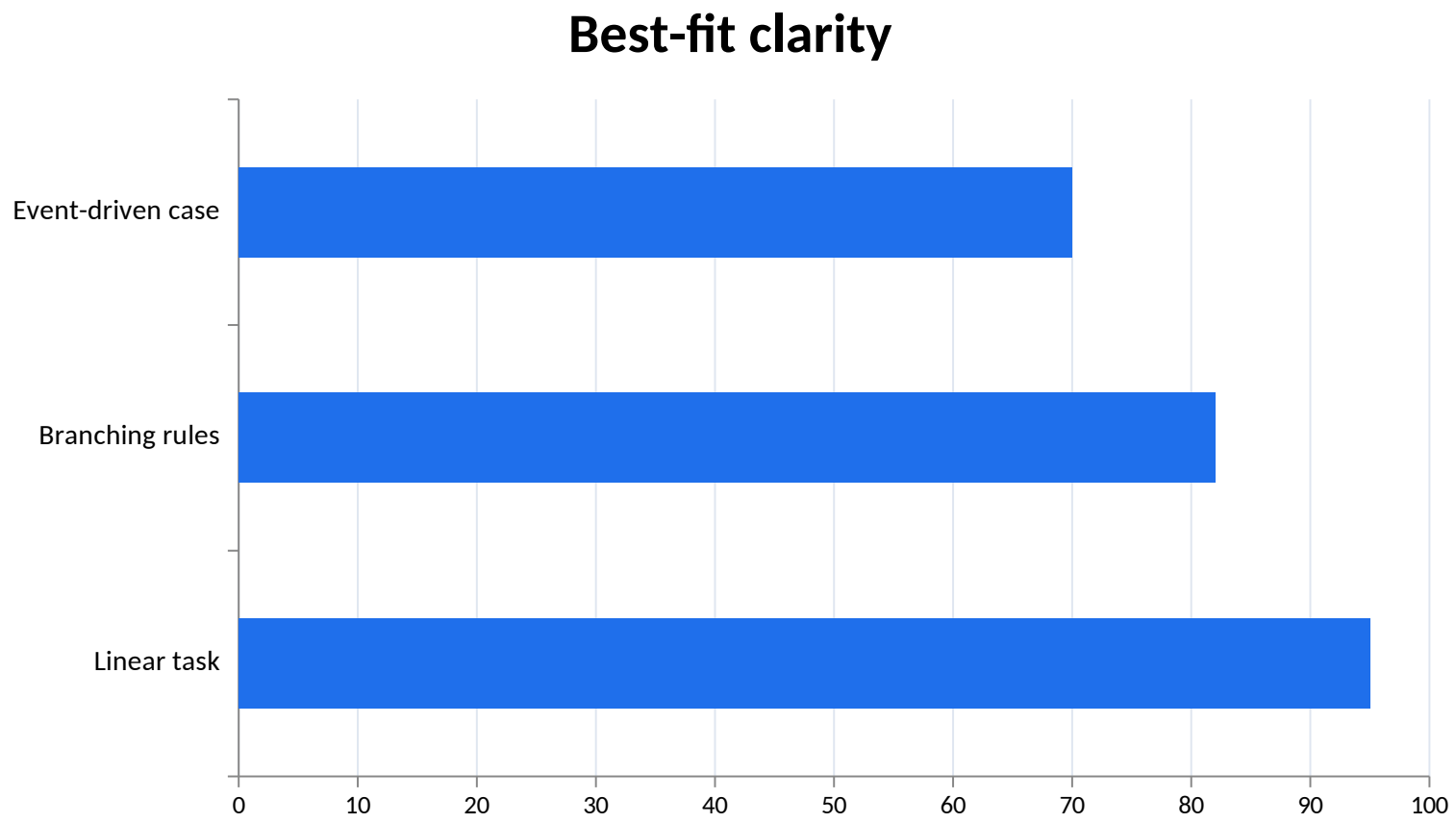
Filter and iterate · Preserve identifiers · Handle blanks

Output

Write Range · CSV or Excel · Reconcile counts

EDITABLE POWERPOINT CHART

Workflow Pattern Decision



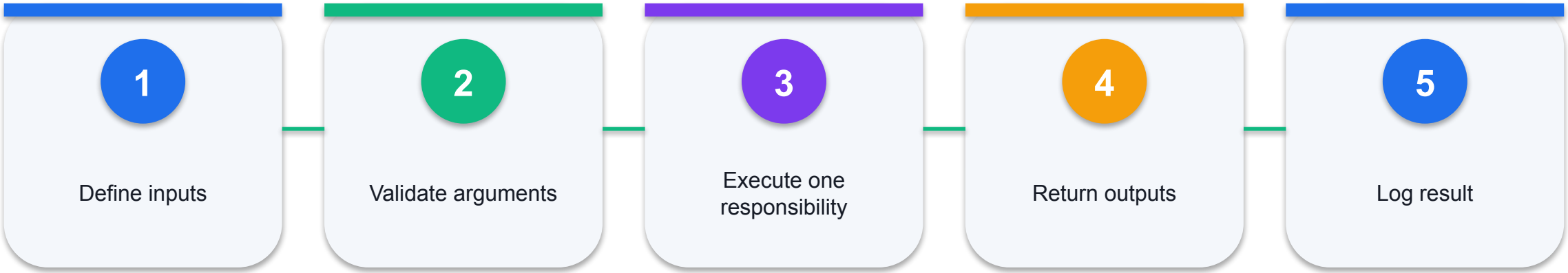
WHAT TO NOTICE

Use the simplest container that makes the process easy to understand. Indicative teaching figures.

Indicative teaching data

WORKFLOW

Reusable Workflow Contract



INSIGHT A reusable workflow should have one clear responsibility and a small, documented contract.

TOPIC 02 · HANDS-ON

Lab 3 · Route Customer Requests with If and Switch

Translate a routing requirement into a readable UiPath flowchart that validates priority and sends each request to the correct queue.

1. Read request
2. Validate priority
3. Evaluate category
4. Set queue
5. Return result

DELIVERABLE A flowchart with validation, If and Switch branches plus a reusable routing output argument

Tools: UiPath Studio Community, Flowchart, If, Switch, arguments · Detailed procedure: Learner Guide and labs/lab-03-*/README.md

EVIDENCE · A2, A3, A4 · K4, K6

Lab 3 · Verification

EXPECTED RESULT

All six requests match ExpectedQueue, including Unknown -> Manual-Review and the two priority routes.

TROUBLESHOOTING

If a category misses its Switch case, normalise the value once before the Switch. If the output argument is empty, ensure the final Assign executes on every path.

EVIDENCE TO RETAIN

Completed route-results workbook and one screenshot of the full flowchart.

TOPIC 02 · HANDS-ON

Lab 4 · Process a Customer DataTable with Loops

Read customer records, validate fields, calculate a service segment and append a traceable result for every row.

1. Read customer table
2. Loop each row
3. Validate fields
4. Assign segment
5. Write result table

DELIVERABLE A For Each Row transaction workflow with row-level validation and an output DataTable

Tools: UiPath Studio Community, Read CSV, DataTable, For Each Row, Add Data Row, Write CSV · Detailed procedure: Learner Guide and labs/lab-04-*/README.md

EVIDENCE · A2, A4 · K4, K6, K11

Lab 4 · Verification

EXPECTED RESULT

The output contains six traceable rows, every segment matches ExpectedSegment, and the blank-email record is routed to Review without stopping the workflow.

TROUBLESHOOTING

If numeric conversion fails, use Double.TryParse or Convert.ToDouble only after checking for blanks. If the output file is locked, close it in Excel before rerunning.

EVIDENCE TO RETAIN

output/customer-segments.csv and a screenshot showing the 6-in / 6-out reconciliation.

CHECK YOUR UNDERSTANDING

Topic 02 · Recap

Sequence

Best for a short linear path with a clear beginning and end.

Flowchart

Makes branching logic and alternate paths easy to inspect.

State machine

Models event-driven work that moves between named states.

Variables hold local state

Name, type, default value and scope determine what a value can do.

Arguments cross boundaries

In, Out and In/Out arguments create explicit contracts between workflows.

Control flow expresses rules

If, Switch, For Each, While and Do While turn requirements into executable logic.

TOPIC 03

03

Automations

UI automation · selectors · recording · browser work · table extraction · Excel integration

LO3 · A4/A5 · K5/K7/K11

Topic 03 · Concept Map

UI elements are targets

A selector describes the application and element a robot should operate.

Reliable selectors use stable attributes

Prefer durable IDs, names and anchors; avoid volatile indexes and session tokens.

Recording accelerates a draft

Recorded actions are a starting point that must be renamed, parameterised and tested.

Table extraction creates structure

Browser rows become a DataTable that can be filtered, validated and written to Excel.

Excel is both source and destination

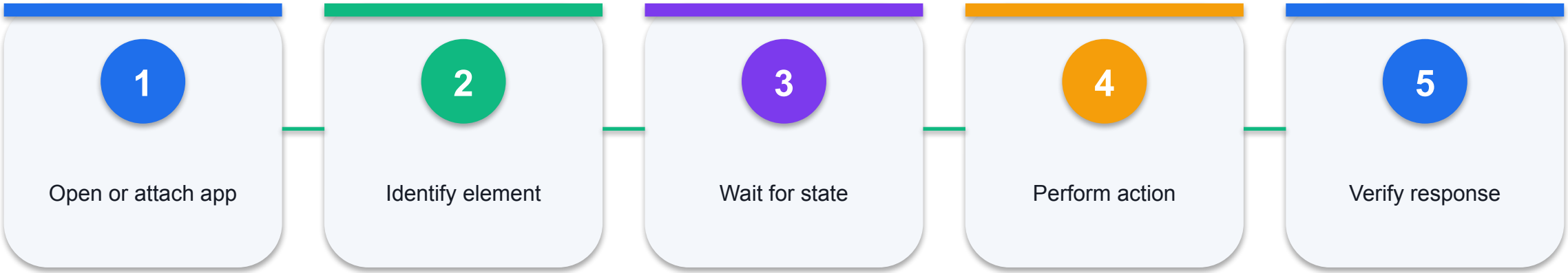
Read Range, Write Range and workbook activities connect business data to workflows.

Validation closes the loop

Counts, required columns, output files and screenshots provide objective evidence.

WORKFLOW

UI Automation Interaction Model



INSIGHT Reliable automation waits for observable states instead of assuming the screen is ready.

PRACTITIONER VIEW

Selector Quality

Stable

Meaningful id or name · Consistent app attributes · Short and specific

Fragile

Dynamic index · Session token · Absolute hierarchy

Recovery

Anchor nearby label · Wildcards with care · Validate after change

DECISION VIEW

Modern vs Classic UI Automation

MODERN

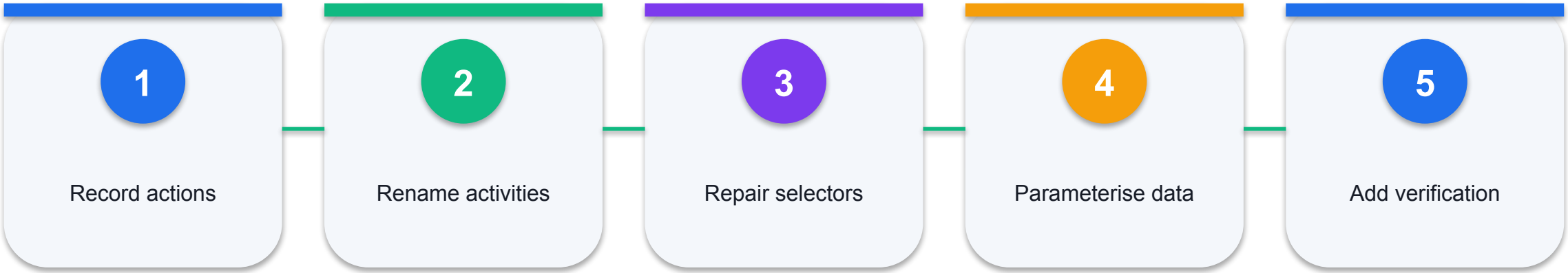
- Application/browser scope
- Unified target and anchor
- Object repository support
- Current design experience

CLASSIC

- Open/Attach activities
- Separate selector properties
- Common in legacy projects
- May require upgrade

WORKFLOW

Recording Is a Drafting Tool



INSIGHT A recording becomes maintainable only after it is cleaned, structured and tested.

PRACTITIONER VIEW

Input Methods

Simulate

Fast and background-capable · No physical mouse · Not supported everywhere

Window messages

Works with many desktop apps · May run in background · Application dependent

Hardware events

Most human-like · Needs foreground · Sensitive to focus

DECISION VIEW

Text Extraction Options

NATIVE TEXT

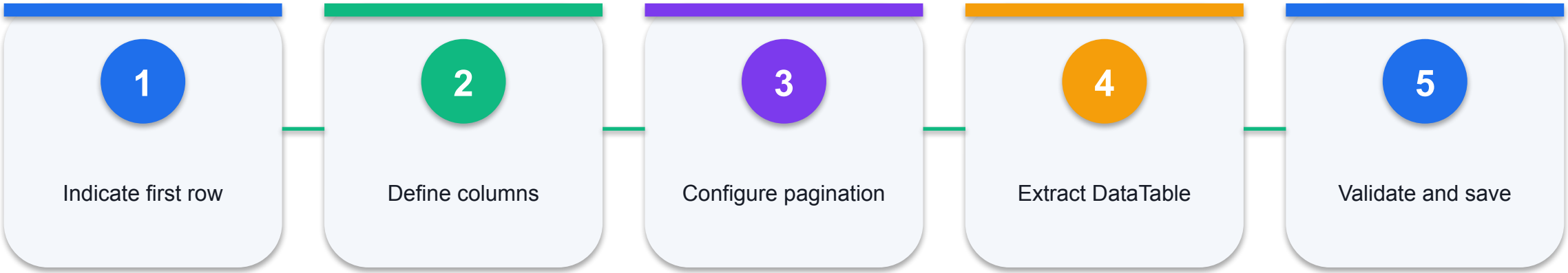
- Get Text for native UI text
- Fast and accurate when accessible
- Depends on application metadata
- Preserves text characters

OCR

- OCR for pixels and scans
- Handles images and remote desktops
- Needs language and confidence checks
- May require human review

WORKFLOW

Structured Table Extraction



INSIGHT The extraction succeeds only when the output schema and record count are verified.

PRACTITIONER VIEW

Web Automation Risks

Page change

Selector no longer matches · Layout changes · A/B tests

Timing

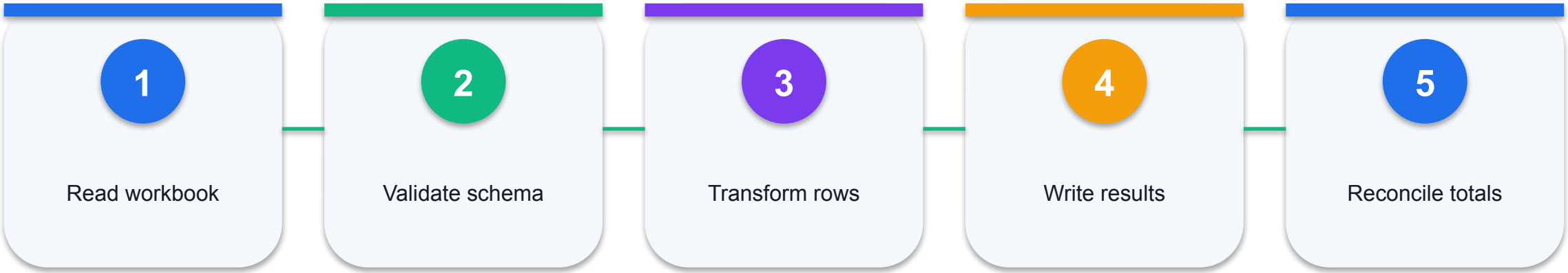
Delayed content · Network latency · Pagination

Access

Authentication · Bot protection · Terms and privacy

WORKFLOW

Excel Automation Pattern



INSIGHT Keep raw input separate from output so the source remains auditable.

DECISION VIEW

Workbook vs Excel Activities

WORKBOOK

- Can work without Excel installed
- Good for basic read/write
- Fast for simple operations
- Limited advanced features

EXCEL

- Uses the Excel application
- Supports richer features
- Needs desktop Excel
- Requires application lifecycle control

PRACTITIONER VIEW

Output Validation

Completeness

Expected row count · Required columns · No missing identifiers

Accuracy

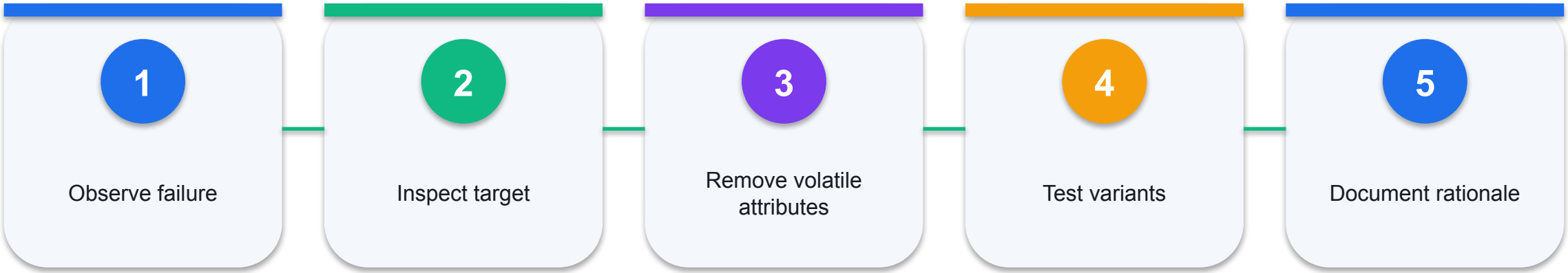
Spot-check values · Types and formats · Business-rule totals

Evidence

Timestamped file · Log summary · Screenshot or result record

WORKFLOW

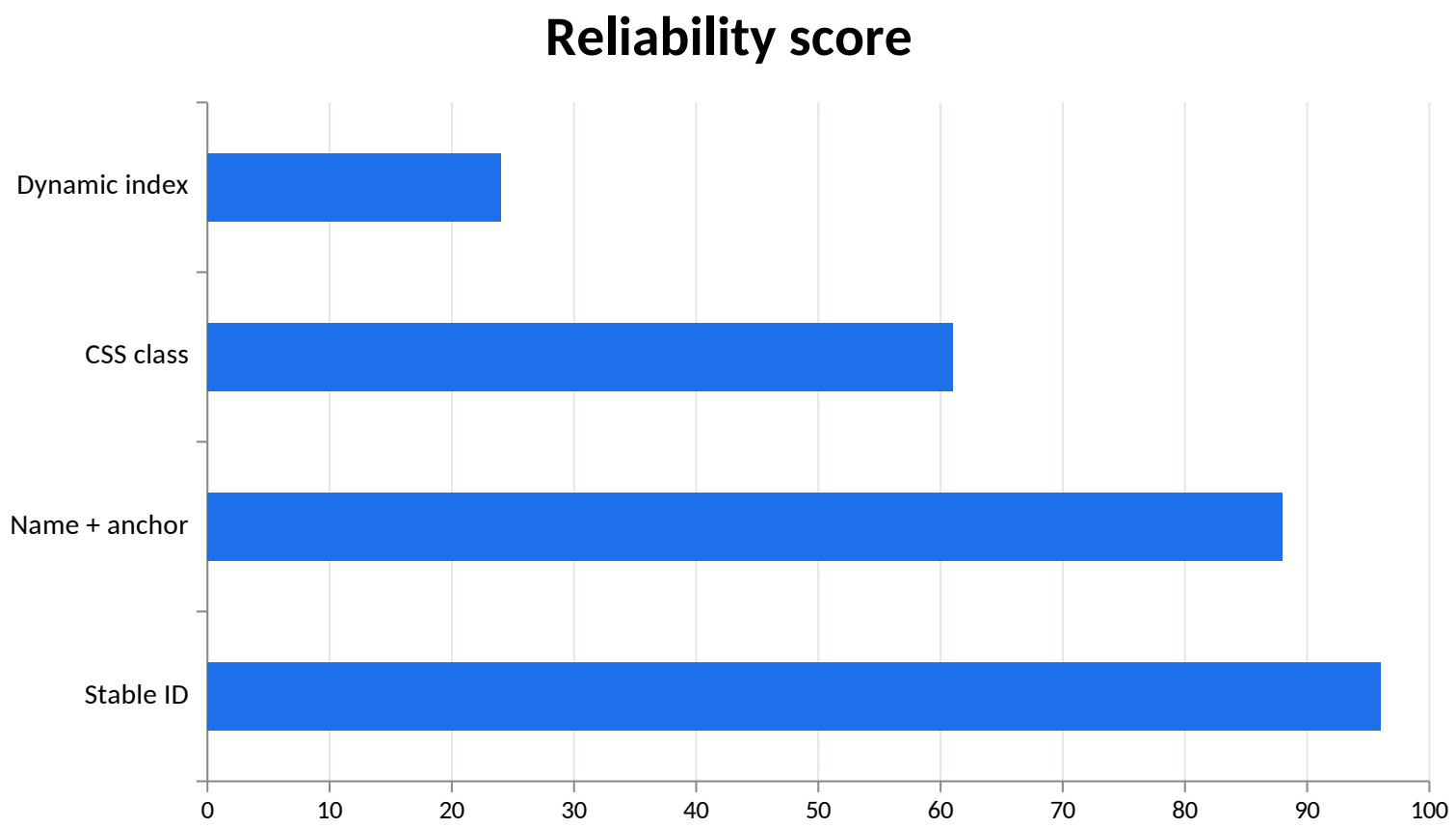
Selector Maintenance Loop



INSIGHT Do not simply add wildcards until a selector works; make each retained attribute intentional.

EDITABLE POWERPOINT CHART

Illustrative Selector Reliability



WHAT TO NOTICE

Stable semantic attributes outperform position-based selectors. Indicative teaching figures.

Indicative teaching data

PRACTITIONER VIEW

Responsible Data Extraction

Permission

Use authorised sources · Respect terms · Avoid bypassing controls

Minimisation

Collect only needed fields · Limit retention · Protect personal data

Quality

Record source and time · Validate schema · Handle missing data

TOPIC 03 · HANDS-ON

Lab 5 · Validate and Reconcile Invoices in Excel

Read an invoice register, validate required values, calculate line totals and write a reconciliation sheet with evidence of exceptions.

1. Open invoice workbook
2. Read input sheet
3. Validate each row
4. Calculate totals
5. Write results and summary

DELIVERABLE An Excel automation that produces Valid, Review and Rejected outcomes with a reconciliation summary

Tools: UiPath Studio Community, Use Excel File, Read Range, For Each Row, Write Range · Detailed procedure: Learner Guide and labs/lab-05-*/README.md

EVIDENCE · A4, A5 · K5, K11

Lab 5 · Verification

EXPECTED RESULT

All six invoices receive the expected status, Valid totals calculate correctly, and the Summary counts reconcile to six input rows.

TROUBLESHOOTING

If Excel remains open after a fault, close the application and rerun with the Excel scope configured to save and close. If dates import as text, treat blank checks separately from display format.

EVIDENCE TO RETAIN

output/invoice-reconciliation.xlsx with Results and Summary sheets.

TOPIC 03 · HANDS-ON

Lab 6 · Extract and Filter a Singapore Schools Table

Use browser table extraction to capture a list of secondary schools, preserve source evidence, filter Ang Mo Kio records and write a validated Excel output.

1. Open authorised page
2. Indicate table
3. Extract records
4. Filter Ang Mo Kio
5. Write and validate output

DELIVERABLE A table-extraction workflow, schools output workbook and Ang Mo Kio count summary

Tools: UiPath Studio Community, Use Application/Browser, Extract Table Data, DataTable, Excel · Detailed procedure: Learner Guide and labs/lab-06-*/README.md

EVIDENCE · A1, A4, A5 · K5, K7

Lab 6 · Verification

EXPECTED RESULT

The deterministic dataset produces six total rows and three Ang Mo Kio rows; the output records the source URL, timestamp and both counts.

TROUBLESHOOTING

If the live table structure changes, use the supplied mock workbook and repair selectors in a separate copy. If the count is zero, trim and normalise PlanningArea before comparing.

EVIDENCE TO RETAIN

output/schools-analysis.xlsx plus screenshots of Extract Table Data and the validated Summary sheet.

CHECK YOUR UNDERSTANDING

Topic 03 · Recap

UI elements are targets

A selector describes the application and element a robot should operate.

Reliable selectors use stable attributes

Prefer durable IDs, names and anchors; avoid volatile indexes and session tokens.

Recording accelerates a draft

Recorded actions are a starting point that must be renamed, parameterised and tested.

Table extraction creates structure

Browser rows become a DataTable that can be filtered, validated and written to Excel.

Excel is both source and destination

Read Range, Write Range and workbook activities connect business data to workflows.

Validation closes the loop

Counts, required columns, output files and screenshots provide objective evidence.

TOPIC 04

04

Debug and Exception Handling

Breakpoints · stepping · panels · logs · Try Catch · retry · evidence · recovery

LO4 · A5/A7/A8 · K8/K13

Topic 04 · Concept Map

Debugging explains behaviour

Breakpoints and stepping reveal values, branches and activity order.

Logs explain operation

Structured messages show what ran, with which transaction and result.

Exceptions need classification

Business exceptions differ from system failures and lead to different responses.

Try Catch contains failure

Catch the specific exception, record evidence and choose retry, skip or stop.

Finally restores state

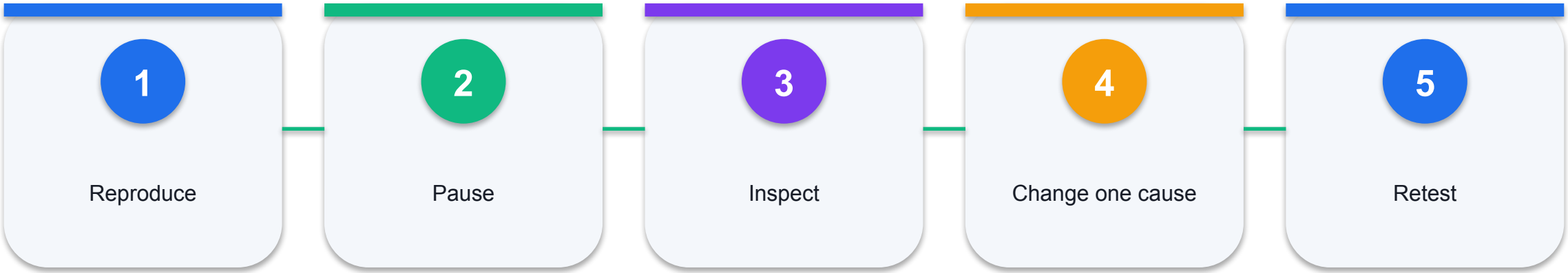
Cleanup actions run whether the Try succeeds or fails.

Evidence supports improvement

Screenshots, logs and transaction summaries turn defects into actionable feedback.

WORKFLOW

Debugging Loop



INSIGHT A controlled debug cycle changes one hypothesis at a time and preserves evidence.

PRACTITIONER VIEW

Debug Actions

Step Into

Enter invoked workflow · Trace detailed behaviour · Use for root cause

Step Over

Run activity as one unit · Stay in current workflow · Use for navigation

Continue

Run to next breakpoint · Check a later state · Avoid blind skipping

DECISION VIEW

Breakpoints vs Slow Step

BREAKPOINT

- Pause at a chosen activity
- Inspect a precise state
- Conditional breakpoints reduce noise
- Repeatable for the same defect

SLOW STEP

- Runs the whole workflow slowly
- Good for visual observation
- Less precise for data defects
- Useful early in a demonstration

PRACTITIONER VIEW

Debug Panels

Locals

Current variables · Arguments and scope · Live values

Watch

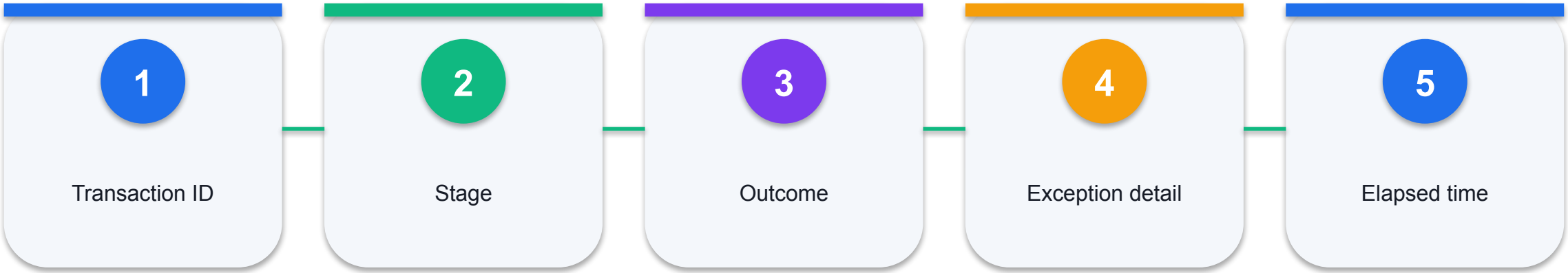
Chosen expressions · Compare expected value · Track across steps

Immediate

Evaluate expression · Test transformation · Do not mutate blindly

WORKFLOW

Structured Logging Pattern



INSIGHT A useful log entry answers which item, which stage, what result and how long.

DECISION VIEW

Business vs System Exception

BUSINESS

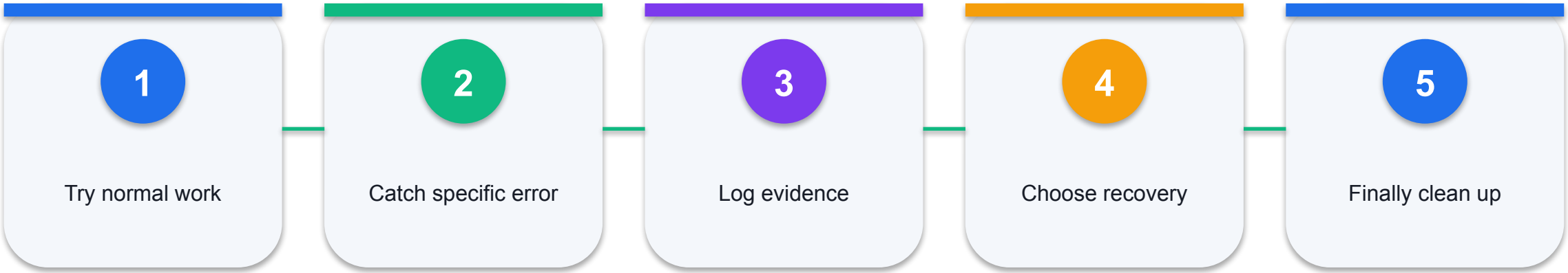
- Input violates a business rule
- Example: missing approval
- Usually skip or route for review
- Retry rarely helps

SYSTEM

- Technology cannot complete action
- Example: timeout or file lock
- Retry may help
- May require technical escalation

WORKFLOW

Try Catch Finally



INSIGHT Catch the narrowest useful exception and preserve enough evidence to diagnose it.

PRACTITIONER VIEW

Retry Design

When

Transient failure · Timeout or temporary lock · Idempotent action

How

Bounded attempts · Delay or backoff · Log each attempt

Stop

Permanent rule failure · Repeated same error · Risk of duplicate action

DECISION VIEW

ContinueOnError Trade-off

POTENTIAL USE

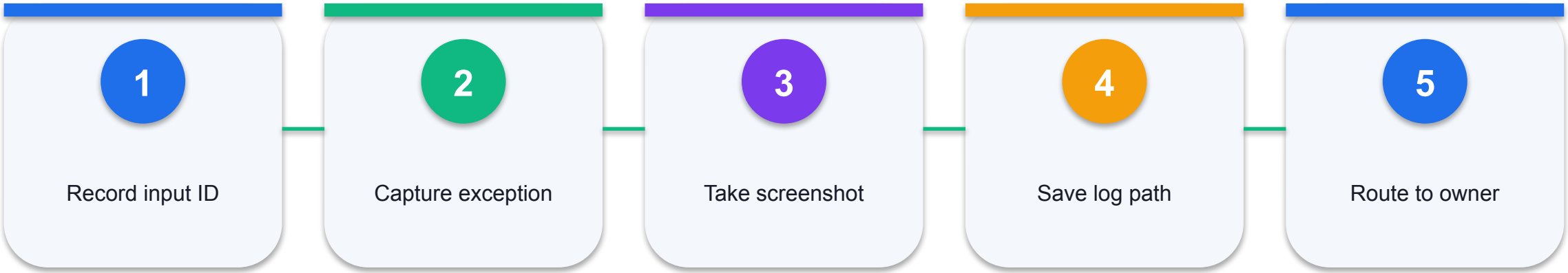
- Keeps the workflow moving
- Can be useful for non-critical cleanup
- May preserve throughput
- Needs explicit result handling

RISK

- Can hide the original failure
- Downstream data may be invalid
- Logs become ambiguous
- Avoid as a blanket fix

WORKFLOW

Evidence Capture



INSIGHT Evidence should be sufficient to reproduce the issue without exposing confidential data.

PRACTITIONER VIEW

Validation Levels

Activity

Correct selector · Expected value · Local test

Workflow

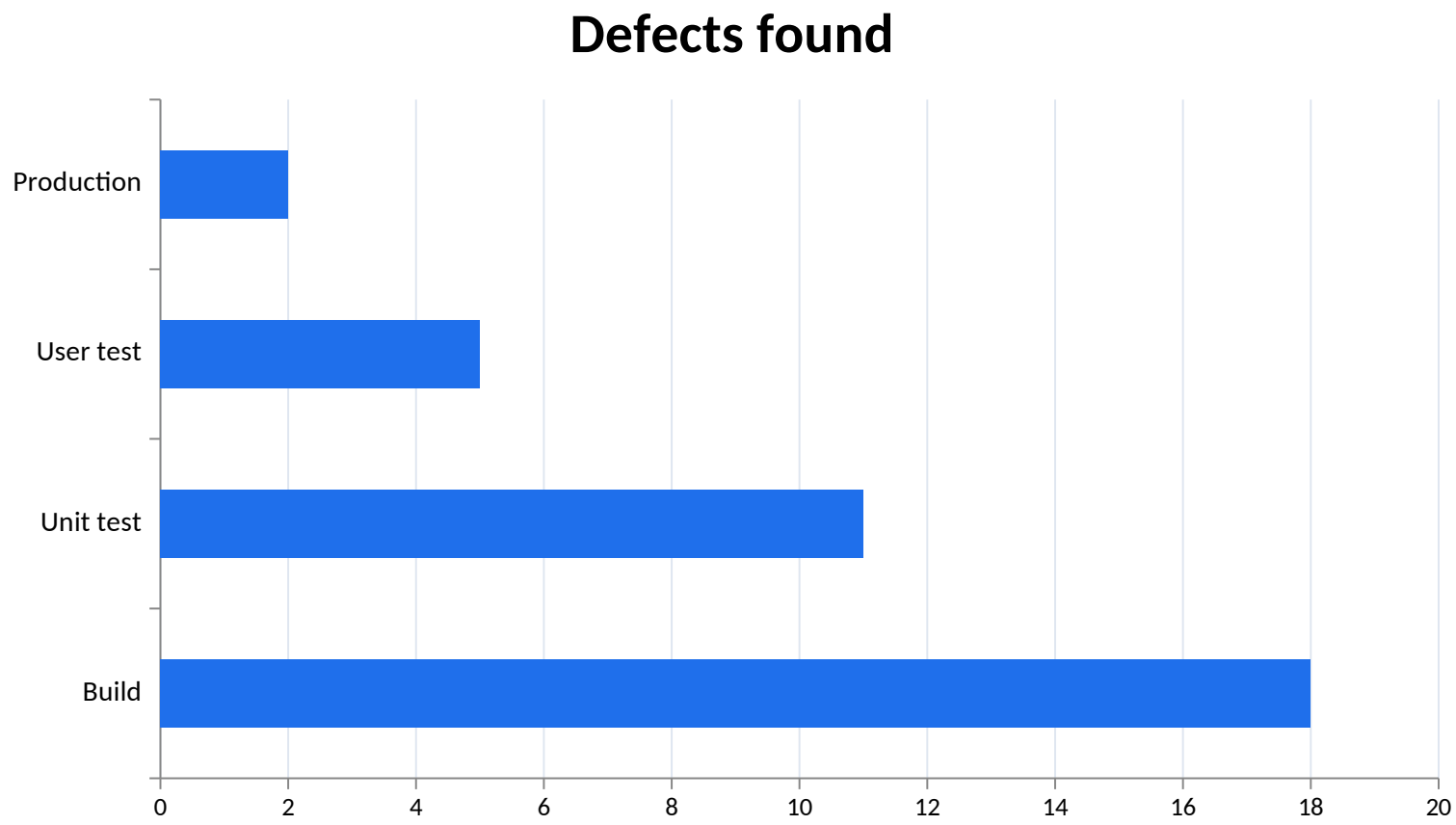
Happy and error paths · Arguments · Output contract

Process

End-to-end data · Business acceptance · Operational monitoring

EDITABLE POWERPOINT CHART

Illustrative Defect Discovery



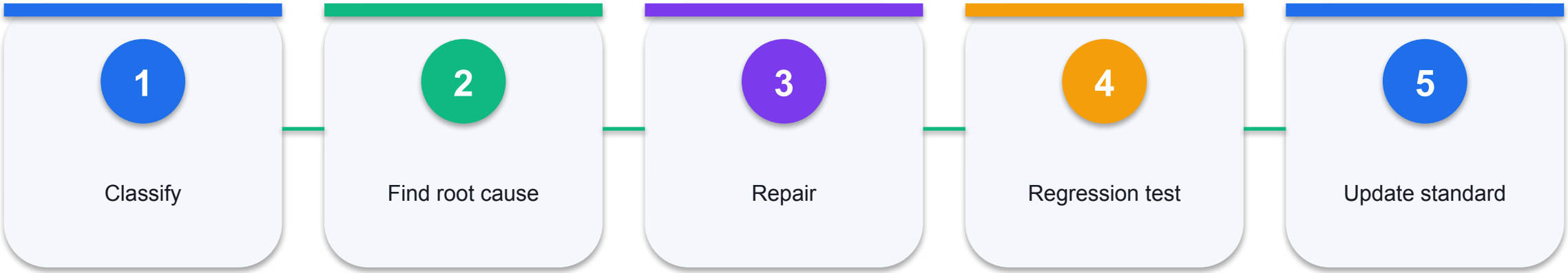
WHAT TO NOTICE

Finding defects earlier makes fixes cheaper and safer. Indicative teaching figures.

Indicative teaching data

WORKFLOW

From Defect to Improvement



INSIGHT A defect is not fully closed until the test and operating standard prevent recurrence.

PRACTITIONER VIEW

Safe Error Messages

Include

Transaction identifier · Failed stage · Recovery action

Exclude

Passwords · Full personal data · Access tokens

Route

Business owner · Support queue · Severity and SLA

TOPIC 04 · HANDS-ON

Lab 7 · Debug a Failing Order Workflow

Use breakpoints, stepping, Locals and Watch to find three deliberate defects in an order-classification workflow, then run regression cases.

1. Reproduce failure
2. Pause at breakpoint
3. Inspect variables
4. Repair one cause
5. Run regression tests

DELIVERABLE A repaired workflow, defect log and regression evidence for six orders

Tools: UiPath Studio Community, breakpoints, Step Into/Over, Locals, Watch, logs · Detailed procedure: Learner Guide and labs/lab-07-*/README.md

EVIDENCE · A5, A7, A8 · K8

Lab 7 · Verification

EXPECTED RESULT

All six cases pass, O5005 is Priority at the boundary, O5006 is Review, and the defect log records symptom, root cause, change and retest evidence.

TROUBLESHOOTING

If the workflow still uses old values, stop Debug rather than rerunning from a paused state. If a Watch expression errors, check variable scope at the paused activity.

EVIDENCE TO RETAIN

Completed defect log and screenshot of a breakpoint with Locals/Watch values.

TOPIC 04 · HANDS-ON

Lab 8 · Build Resilient File Processing with Try Catch

Process a file manifest, classify missing or invalid files, retry transient access failures and produce a clean exception summary without exposing sensitive data.

1. Read manifest
2. Try file action
3. Classify exception
4. Retry or route
5. Write secure summary

DELIVERABLE A Try Catch workflow with bounded retry, Finally cleanup and a structured exception report

Tools: UiPath Studio Community, Try Catch, Retry Scope, Log Message, file activities · Detailed procedure: Learner Guide and labs/lab-08-*/README.md

EVIDENCE · A5, A7, A8 · K8, K13

Lab 8 · Verification

EXPECTED RESULT

All six rows produce one final result, retries never exceed three, business and system exceptions are distinct, and no sensitive value appears in logs.

TROUBLESHOOTING

If the workflow continues after a critical system failure, check ContinueOnError and the Catch recovery path. If results duplicate, append only once in Finally.

EVIDENCE TO RETAIN

output/exception-summary.csv and screenshots of Try Catch plus one retry log sequence.

CHECK YOUR UNDERSTANDING

Topic 04 · Recap

Debugging explains behaviour

Breakpoints and stepping reveal values, branches and activity order.

Logs explain operation

Structured messages show what ran, with which transaction and result.

Exceptions need classification

Business exceptions differ from system failures and lead to different responses.

Try Catch contains failure

Catch the specific exception, record evidence and choose retry, skip or stop.

Finally restores state

Cleanup actions run whether the Try succeeds or fails.

Evidence supports improvement

Screenshots, logs and transaction summaries turn defects into actionable feedback.

TOPIC 05

05

Manage RPA

Packages · processes · machines · runtimes · jobs · triggers · Assistant · monitoring · optimisation

LO5 · A7/A8/A9 · K9/K12/K13

Topic 05 · Concept Map

Studio creates packages

Publishing turns a tested project into a versioned automation package.

Orchestrator creates processes

A process associates a package version with a folder and runtime context.

Machines provide execution capacity

Machine templates and runtimes connect infrastructure to folders.

Jobs are executions

Start now, schedule with triggers, or launch an attended process through Assistant.

Monitoring reveals health

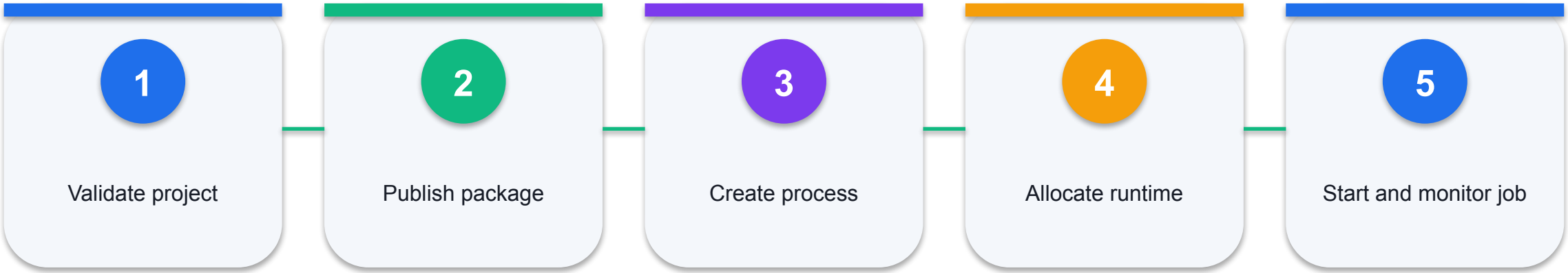
Statuses, logs, queues and alerts expose throughput, exceptions and bottlenecks.

Optimisation is evidence-led

Measure cycle time, success rate and exception causes before changing the robot.

WORKFLOW

Studio to Orchestrator Lifecycle



INSIGHT Deployment is a versioned chain from source project to package, process and job.

PRACTITIONER VIEW

Orchestrator Resource Model

Tenant and folder

Isolation boundary · Users and permissions · Resources grouped

Machine and runtime

Execution capacity · Attended/unattended · Folder assignment

Automation

Package and process · Jobs and triggers · Logs and queues

DECISION VIEW

Package vs Process

PACKAGE

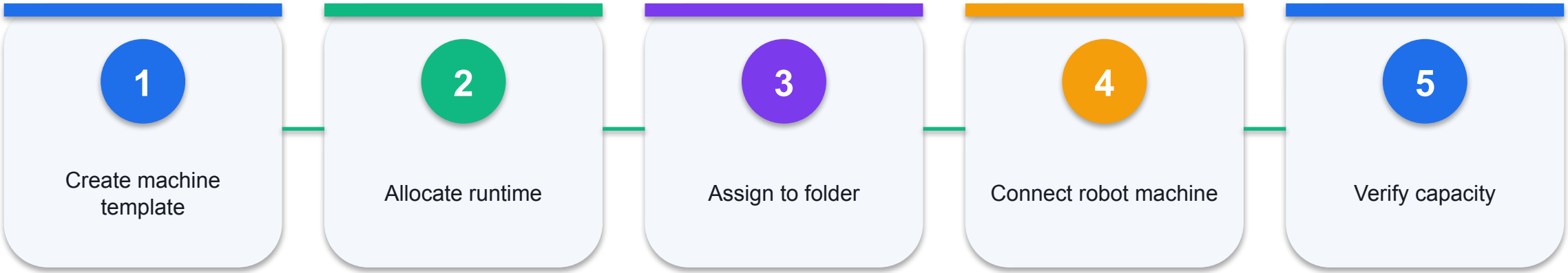
- Versioned output from Studio
- Contains workflows and dependencies
- Stored in a feed
- Immutable release unit

PROCESS

- Deployable configuration
- Selects a package version
- Lives in a folder
- Used to start jobs

WORKFLOW

Machine Connection Model



INSIGHT Connection settings prove identity; folder and runtime assignments grant usable execution capacity.

DECISION VIEW

Attended vs Unattended Runtime

ATTENDED

- Signed-in user context
- Launched through Assistant
- Supports interactive work
- User credentials remain central

UNATTENDED

- Managed robot account
- Started by job or trigger
- Processes queues at scale
- Credential and machine governance required

PRACTITIONER VIEW

Jobs and Triggers

Job

One process execution · Pending/running/faulted · Input and output arguments

Time trigger

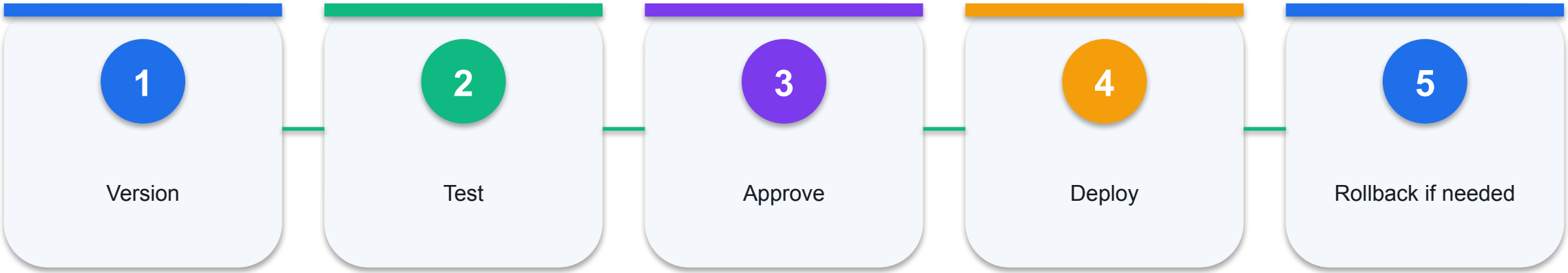
Calendar schedule · Business-hour windows · Avoid overlaps

Queue trigger

Starts from workload · Scales with transactions · Requires queue design

WORKFLOW

Release Control



INSIGHT Keep the previous proven package available and document which version each environment runs.

PRACTITIONER VIEW

Orchestrator Security

Identity

Named users and robot accounts · Least privilege · Multi-factor controls

Secrets

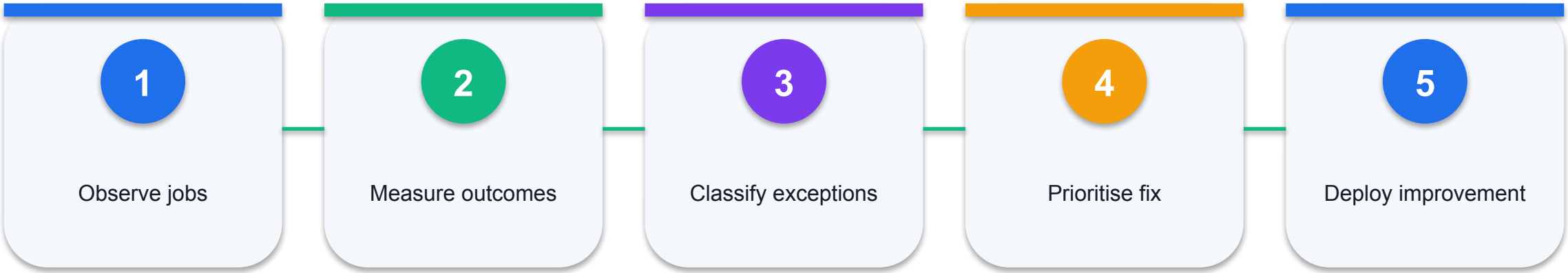
Assets and credential stores · Never in XAML or logs · Rotate access

Audit

Folder roles · Deployment history · Job and user events

WORKFLOW

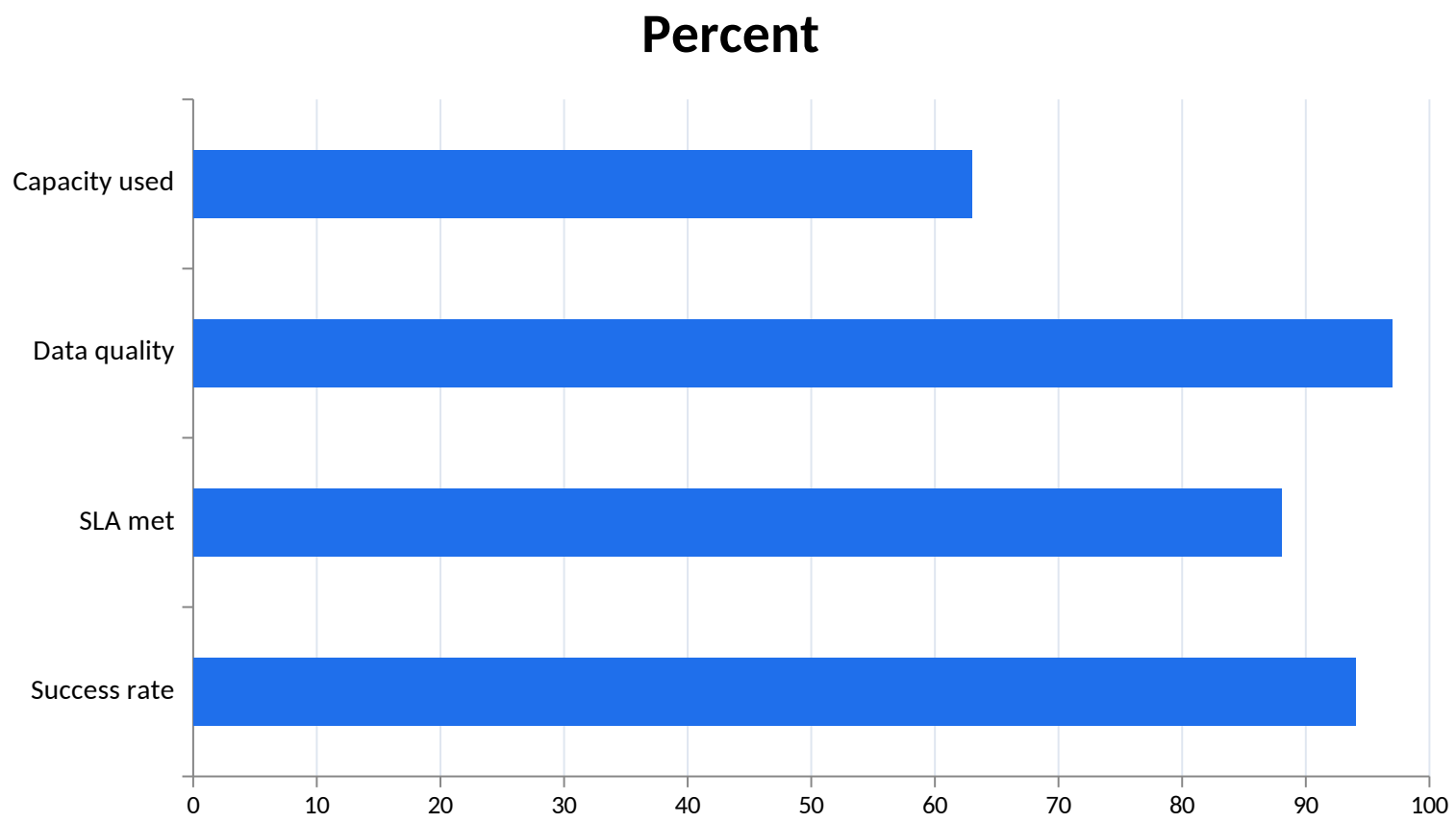
Monitoring Feedback Loop



INSIGHT Operational data should drive optimisation, not intuition alone.

EDITABLE POWERPOINT CHART

Illustrative Robot Health Dashboard



WHAT TO NOTICE

Health combines quality, timeliness and capacity. Indicative teaching figures.

Indicative teaching data

PRACTITIONER VIEW

Operational Measures

Reliability

Success rate · Exception rate · Mean time to recover

Performance

Cycle time · Queue age · Throughput

Value

Hours released · Avoided rework · Service outcome

DECISION VIEW

Scale Up vs Scale Out

SCALE UP

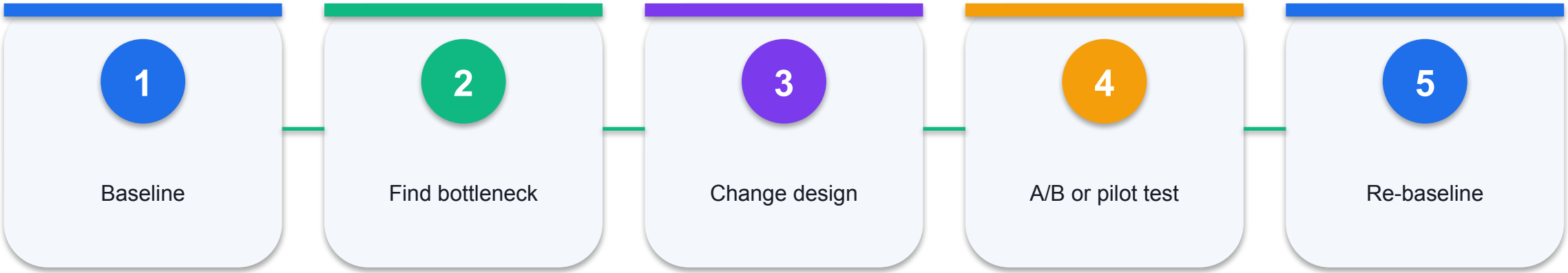
- Faster or larger machine
- Simple architecture
- Limited parallelism
- Single-capacity risk

SCALE OUT

- More runtimes or machines
- Parallel job execution
- Better resilience
- Needs queue and licence planning

WORKFLOW

Optimisation Cycle



INSIGHT Change one measurable bottleneck and prove the new version is better before scaling it.

PRACTITIONER VIEW

Production Readiness Gate

Build

Named activities · Configuration externalised · Dependencies pinned

Test

Happy and exception paths · Evidence captured · Business acceptance

Operate

Owner and support model · Monitoring and alerts · Rollback plan

Lab 9 · Publish and Run a Process with Orchestrator

Publish a validated project, create or update an Orchestrator process, start a job or Assistant run, and capture version and execution evidence.

1. Validate project
2. Publish package
3. Create process
4. Start job
5. Review logs and version

DELIVERABLE A published package, deployed process, successful job and release evidence record

Tools: UiPath Studio Community, Automation Cloud Orchestrator, UiPath Assistant · Detailed procedure: Learner Guide and labs/lab-09-*/README.md

EVIDENCE · A7, A8, A9 · K9, K12

Lab 9 · Verification

EXPECTED RESULT

The package version is traceable to a deployed process and a successful execution, with no credential or machine-key exposure in evidence.

TROUBLESHOOTING

If the package is not visible, verify tenant, folder and feed. If a job stays Pending, check machine/runtime allocation and folder assignment rather than repeatedly starting more jobs.

EVIDENCE TO RETAIN

Completed release record plus screenshots of the process version and successful job details.

TOPIC 05 · HANDS-ON

Lab 10 · Design and Monitor a Multi-Robot Queue

Model a dispatcher-worker pattern, calculate queue metrics and decide when to scale, retry or route work for human review.

1. Load transactions
2. Dispatch queue items
3. Workers claim items
4. Record outcomes
5. Optimise capacity

DELIVERABLE A dispatcher-worker UiPath design, queue simulation workbook and evidence-led optimisation recommendation

Tools: UiPath Studio Community, Orchestrator queues, dispatcher/worker pattern, monitoring · Detailed procedure: Learner Guide and labs/lab-10-*/README.md

EVIDENCE · A1, A2, A3, A4, A5, A7, A8, A9 · K3, K9, K13

Lab 10 · Verification

EXPECTED RESULT

Every transaction has a unique reference and final state, retries are bounded, business exceptions are not retried, and the optimisation recommendation cites measured queue evidence.

TROUBLESHOOTING

If workers duplicate items, verify unique references and transaction-state updates. If queue age grows while success rate falls, fix the dominant exception before adding capacity.

EVIDENCE TO RETAIN

Queue metrics workbook, workflow screenshots and a written scale/retry/review recommendation.

CHECK YOUR UNDERSTANDING

Topic 05 · Recap

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Monitoring reveals health

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Optimisation is evidence-led

Measure cycle time, success rate and exception causes before changing the robot.

TOPIC ✓

Course Wrap-Up

From opportunity selection to a monitored, resilient RPA process

OUTCOMES ACHIEVED

What You Can Now Do

LO1

Determine appropriate RPA opportunities and best practices for attended and unattended automation.

LO2

Plan and construct UiPath workflows using sequences, flowcharts, state machines, variables, arguments and control flow.

LO3

Design and validate UI, web, data and Excel automation processes with reliable selectors and structured outputs.

LO4

Test, debug, log and handle exceptions in RPA workflows.

LO5

Package, deploy, monitor and optimise UiPath processes using Assistant and Orchestrator.

KEEP LEARNING

Recommended Follow-On Courses

1. Business Process Automation with Power Automate and Copilot Studio Agents
2. Agentic AI Automation with n8n
3. Python Programming for Beginners
4. Data Analytics with Microsoft Excel
5. Robotics and Automation Fundamentals

FINAL CHECK

Assessment readiness

Use the deck for concepts, the Learner Guide for procedures, and your completed lab evidence for practical rehearsal. Complete both components independently.

AFTER CLASS

Support

Email

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Telephone

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Website

www.tertiarycourses.com.sg

Lab repository

Public learner files and UiPath lab projects

KEEP AUTOMATING

Thank you

You are ready to identify, build, validate and manage beginner-level RPA solutions with UiPath.